

Smart Stick for the Visually Impaired: Final Project Report

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ABSTRACT

The Smart Stick for the Blind is a mobility aid designed to assist visually impaired individuals in safely navigating their environment by detecting obstacles and environmental hazards. Using an ultrasonic sensor for obstacle detection and a raindrop sensor for identifying water or mud, the device provides real-time alerts through a buzzer and vibrator. Controlled by an ATmega328P microcontroller, this project achieves accurate, timely feedback while maintaining efficient power usage with a 9V battery and 7805 - voltage regulator. This report details the device's design, implementation, and potential future enhancements.

INTRODUCTION

Navigating independently poses a significant challenge for visually impaired individuals, especially in complex or unfamiliar environments. Standard mobility aids, such as white canes, often provide limited feedback, focusing primarily on direct obstacles. Motivated by the need for enhanced situational awareness and independence, the **Smart Stick for the Blind** was designed to detect obstacles and hazardous surfaces, such as wet or muddy ground, and to provide immediate alerts.

The device integrates an ultrasonic sensor to detect obstacles up to 50 cm away and a raindrop sensor to identify environmental hazards. Feedback is delivered through a buzzer and a vibrator motor, ensuring both audio and tactile cues. The ATmega328P microcontroller, chosen for its low power consumption and versatile features, manages the sensors and alerts. This report documents the methodology, performance, and potential for future improvements in the Smart Stick's design.

OBJECTIVES

The device's main objectives are:

- **Obstacle Detection:** Use ultrasonic sensing to detect obstacles within 50 cm.
- **Environmental Hazard Detection:** Detect water or mud on the ground using a raindrop sensor.
- **Dual Sensory Alerts:** Provide auditory and tactile feedback to the user.
- **Long Battery Life:** Ensure up to 12 hours of operation on a 9V battery.
- **User-Friendly Design:** Build a lightweight, easy-to-use, and durable device.

THEORY AND DESIGN

Microcontroller Overview: ATmega328P

The **ATmega328P** microcontroller serves as the core of the device, chosen for its low power consumption, flexibility, and comprehensive features:

- **Clock Speed:** Operates at up to 16 MHz, sufficient for real-time data processing.
- **I/O Ports:** Provides 23 programmable I/O lines, used to interface with sensors, buzzer, and vibrator.
- **ADC (Analog-to-Digital Converter):** Contains an 8-channel 10-bit ADC, ideal for reading the analog output from the raindrop sensor.
- **Memory:** Equipped with 32KB flash memory for program storage, 1KB EEPROM, and 2KB SRAM for data handling.

Sensors Used

- **Ultrasonic Sensor:**
 - Works by emitting sound waves and measuring the time taken for the echo to return. The ATmega328P's I/O ports manage the trigger and echo pins. Distance d to an obstacle is calculated by:

$$d = \frac{\text{Speed of sound} \times \text{Echo time}}{2}$$

- In the code, the timeout is set at 2916 μs for a 50 cm distance threshold.
- **Raindrop Sensor:**
 - Detects water presence by sensing conductivity between two pins when exposed to water or mud. The analog value read by the ATmega328P's ADC determines if an alert should be triggered, with a threshold of 900.

Power Management

The Smart Stick for the Blind is powered by a **9V battery** to ensure portability and ease of use. To safely power the ATmega328P microcontroller and connected components, the device uses a **7805-voltage regulator**. This regulator steps down the 9V battery input to a stable **5V output**, which is compatible with the ATmega328P and other 5V components like the ultrasonic sensor and buzzer.

The 7805-voltage regulator not only converts the voltage but also helps protect the microcontroller and sensors from potential fluctuations in the battery's output. This regulated 5V supply ensures reliable and consistent performance for the device, maintaining stable operation over extended usage. Given the microcontroller's low power consumption and the 7805 efficient regulation, the device can operate continuously for over 12 hours on a single 9V battery.

METHODOLOGY

Component Connections and Pin Configuration

- **Ultrasonic Sensor:** Connected to **PB1** (**TRIG_PIN**) for the trigger pulse and **PB0** (**ECHO_PIN**) to receive the echo pulse.
- **Raindrop Sensor:** Connected to **PC0**, using the **ADC0** channel for reading analog values.
- **Alert System:** Buzzer and vibrator connected to **PB2** and **PB3**, configured as output pins for activating alerts.

Microcontroller Configuration and Code Implementation

- **GPIO Configuration (Data Direction and PORT Registers)**

The Data Direction Register B (**DDRB**) configures the direction of each pin:

```
DDRB |= (1 << BUZZER_PIN) | (1 << VIBRATOR_PIN) | (1 << TRIG_PIN); // Set as outputs
DDRB &= ~(1 << ECHO_PIN); // Set as input
```

The **PORTB** register controls output states for the alert components:

- **PORTB** | = (1 << **BUZZER_PIN**); turns the buzzer on, while **PORTB** & = ~ (1 << **BUZZER_PIN**); turns it off
- **ADC Configuration for Raindrop Sensor (ADMUX and ADCSRA Registers)**

ADMUX Register: Configures the ADC reference voltage and input channel:

```
ADMUX |= (1 << REFS0);
ADMUX = (ADMUX & 0xF0) | (RAIN_SENSOR_PIN & 0x0F);
```

ADCSRA Register: Controls the ADC prescaler and enables ADC:

```
ADCSRA |= (1 << ADPS2) | (1 << ADPS1) | (1 << ADPS0);
ADCSRA |= (1 << ADEN);
```

Ultrasonic Sensor Pulse Timing

1. **Trigger Pulse:** Sets **TRIG_PIN** high for 10 μ s, sending a pulse from the ultrasonic sensor.
2. **Echo Pulse Measurement:**
 - The program waits for an echo using **PINB** & (1 << **ECHO_PIN**) and counts microseconds to measure the echo time.
 - This time is then converted into a distance measurement using:

$$Distance = \frac{Echo\ time}{58}$$

Control Logic

In the main loop, the ADC value from the raindrop sensor is compared to ***RAIN_THRESHOLD***. If the value is below this threshold, the buzzer and vibrator are activated. Similarly, the ultrasonic sensor distance reading is compared to ***DISTANCE_THRESHOLD***, activating alerts if an obstacle is detected.

```
// Control buzzer and vibrator based on sensor readings
if (rainDetected || obstacleDetected) {
    if (rainDetected) {
        buzzerOn();    // Turn on buzzer when rain is detected
        vibratorOn();  // Turn on vibrator when rain is detected
    }
    if (obstacleDetected) {
        buzzerOn();    // Turn on buzzer when obstacle is detected
        vibratorOn();  // Turn on vibrator when obstacle is detected
    }
} else {
    buzzerOff();    // Turn off buzzer when no rain or obstacle
    vibratorOff();  // Turn off vibrator when no rain or obstacle
}

_delay_ms(10); // Wait for 10 milliseconds before the next reading
}
```

The delay function ***_delay_ms(10)***; at the end of the loop prevents excessive power consumption while ensuring responsiveness.

RESULTS AND ANALYSIS

The Smart Stick for the Blind was tested extensively in both controlled and real-world environments to evaluate its obstacle detection, hazard identification, and power efficiency. Overall, the device performed as expected, meeting the primary objectives of accuracy, reliability, and usability.

1. **Obstacle Detection:** The ultrasonic sensor reliably detected obstacles within the 50 cm range, providing timely alerts through the buzzer and vibrator. The device responded instantly, with minimal delay in activating the alert system, which is essential for navigating safely in real time. During testing in crowded areas, the device maintained its accuracy, successfully differentiating between nearby obstacles and open spaces.

2. **Environmental Hazard Detection:** The raindrop sensor accurately identified wet surfaces and muddy areas, triggering an alert when the threshold was crossed. This feature proved effective in environments with different types of surfaces, such as wet pavements and muddy paths. The sensitivity of the raindrop sensor was suitable, and it provided feedback without false positives, ensuring reliable performance.
3. **Power Efficiency and Durability:** Powered by a 9V battery regulated to 5V by a 7805-voltage regulator, the device operated continuously for over 12 hours. This confirmed that the power management system is efficient, making the device suitable for daily use. Additionally, the PVC casing and lightweight design contributed to durability and ease of handling.
4. **User Feedback and Practicality:** Feedback from users indicated that the dual-sensory feedback system was valuable in noisy environments, where only a sound alert might be insufficient. Users found the vibration alert particularly useful, as it provided an immediate physical response when obstacles or hazards were detected.
5. **Areas for Improvement:** Testing suggested several potential enhancements to improve the device's versatility. For example, integrating multiple ultrasonic sensors could expand the device's detection range, allowing users to detect obstacles in a wider field of view. Additionally, implementing a rechargeable battery system would enhance sustainability and reduce maintenance costs, aligning with user preferences for a low-maintenance device.

DISCUSSION

The Smart Stick for the Blind performed effectively in real-world testing, providing accurate and timely alerts for obstacles and environmental hazards. The dual-sensory feedback, consisting of audible and tactile alerts, proved particularly useful in environments where sound alone may not be sufficient. The integration of the ultrasonic and raindrop sensors with the ATmega328P microcontroller ensured reliable detection and minimal false positives.

Potential Modifications: To further enhance the device's functionality, the following modifications could be considered:

1. **Extended Range with Multiple Sensors:** Adding additional ultrasonic sensors could increase the detection range and widen the field of view. This modification could help users navigate more complex environments by detecting obstacles from multiple directions.
2. **Bluetooth Integration for Connectivity:** Adding a Bluetooth module could allow the device to send data to a smartphone, enabling location-based alerts, historical data tracking, or integration with navigation apps. This would further enhance the independence and convenience for the user.

3. **Rechargeable Battery System:** Replacing the 9V battery with a rechargeable battery system would improve sustainability and reduce long-term costs. It could also be paired with an indicator to notify the user when the battery is low.

These modifications could further increase the functionality and usability of the Smart Stick, expanding its usefulness in varied environments and providing additional conveniences for visually impaired users.

CONCLUSION

The Smart Stick for the Blind has successfully met its design objectives, proving to be a reliable and effective mobility aid for visually impaired individuals. The device functioned as expected, providing real-time alerts through the buzzer and vibrator whenever an obstacle or hazardous surface was detected. Testing demonstrated that the ultrasonic sensor consistently detected obstacles within the designated 50 cm range, while the raindrop sensor accurately identified water or muddy areas, triggering the alert system without significant delays.

The project was successful in achieving its primary goals, including accurate obstacle detection, effective hazard identification, and dual-sensory feedback. Additionally, the power management system, comprising a 9V battery and 7805 voltage regulator, allowed the device to operate continuously for over 12 hours, confirming its suitability for daily use. The device's compact and ergonomic design further enhances its practicality and ease of use for visually impaired individuals.

Overall, the Smart Stick for the Blind is a dependable, user-friendly solution that addresses a vital need for improved mobility and independence among visually impaired users. This success demonstrates the potential impact of accessible technology in improving quality of life for those with visual impairments.

APPENDIX

```
#include <avr/io.h>
#include <util/delay.h>
#include <stdbool.h> // Include this for the bool type

#define F_CPU 16000000UL // Clock speed
#define BUZZER_PIN PB2    // Define buzzer pin
#define VIBRATOR_PIN PB3  // Define vibrator pin
#define TRIG_PIN PB1      // Define trigger pin for ultrasonic sensor
#define ECHO_PIN PB0      // Define echo pin for ultrasonic sensor
#define RAIN_SENSOR_PIN PC0 // Define raindrop sensor analog pin
#define RAIN_THRESHOLD 900 // Rain threshold value for the buzzer and vibrator
```

```

#define DISTANCE_THRESHOLD 50 // Distance threshold for the ultrasonic sensor in
cm
#define TIMEOUT 2915 // Timeout for echo signal (50 cm / 343 m/s)

// Function prototypes
void initBuzzer(void);
void buzzerOn(void);
void buzzerOff(void);
void initVibrator(void);
void vibratorOn(void);
void vibratorOff(void);
void initUltrasonicSensor(void);
void initRaindropSensor(void);
uint16_t readRaindropSensor(void);
unsigned int getDistance(void);

void initBuzzer(void) {
    DDRB |= (1 << BUZZER_PIN); // Set buzzer pin as output
}

void buzzerOn(void) {
    PORTB |= (1 << BUZZER_PIN); // Turn buzzer on
}

void buzzerOff(void) {
    PORTB &= ~(1 << BUZZER_PIN); // Turn buzzer off
}

void initVibrator(void) {
    DDRB |= (1 << VIBRATOR_PIN); // Set vibrator pin as output
}

void vibratorOn(void) {
    PORTB |= (1 << VIBRATOR_PIN); // Turn vibrator on
}

void vibratorOff(void) {
    PORTB &= ~(1 << VIBRATOR_PIN); // Turn vibrator off
}

void initUltrasonicSensor(void) {
    DDRB |= (1 << TRIG_PIN); // Set trigger pin as output
    DDRB &= ~(1 << ECHO_PIN); // Set echo pin as input
}

```



```

void initRaindropSensor(void) {
    // Set the ADC prescaler to 128 (for 16MHz clock)
    ADCSRA |= (1 << ADPS2) | (1 << ADPS1) | (1 << ADPS0);
    // Set reference voltage to AVcc with external capacitor at AREF pin
    ADMUX |= (1 << REFS0);
    // Enable ADC
    ADCSRA |= (1 << ADEN);
}

```

```

uint16_t readRaindropSensor(void) {
    // Set the appropriate channel (rain sensor on PC0)
    ADMUX = (ADMUX & 0xF0) | (RAIN_SENSOR_PIN & 0x0F);
    // Start conversion
    ADCSRA |= (1 << ADSC);
    // Wait for conversion to complete
    while (ADCSRA & (1 << ADSC));
    // Return the analog value
    return ADC;
}

```

```

unsigned int getDistance(void) {
    uint32_t count = 0;

    // Send trigger pulse
    PORTB &= ~(1 << TRIG_PIN);
    _delay_us(2);
    PORTB |= (1 << TRIG_PIN);
    _delay_us(10);
    PORTB &= ~(1 << TRIG_PIN);

    // Wait for echo start (with timeout)
    count = 0;
    while (!(PINB & (1 << ECHO_PIN))) {
        if (count++ > TIMEOUT) return DISTANCE_THRESHOLD; // Timeout
        _delay_us(1);
    }

    // Measure echo duration (with timeout)
    count = 0;
    while (PINB & (1 << ECHO_PIN)) {
        if (count++ > TIMEOUT) return DISTANCE_THRESHOLD; // Timeout
        _delay_us(1);
    }

    // Calculate distance in cm

```

```

    return (count / 58); // Speed of sound = 343 m/s
}

int main(void) {
    uint16_t rainValue;
    unsigned int distance;

    initBuzzer();
    initVibrator();
    initUltrasonicSensor();
    initRaindropSensor();

    while (1) {
        // Read sensor values
        rainValue = readRaindropSensor();
        distance = getDistance();

        // Determine if sensors are triggered
        bool rainDetected = (rainValue <= RAIN_THRESHOLD);
        bool obstacleDetected = (distance < DISTANCE_THRESHOLD);

        // Control buzzer and vibrator based on sensor readings
        if (rainDetected || obstacleDetected) {
            if (rainDetected) {
                buzzerOn(); // Turn on buzzer when rain is detected
                vibratorOn(); // Turn on vibrator when rain is detected
            }
            if (obstacleDetected) {
                buzzerOn(); // Turn on buzzer when obstacle is detected
                vibratorOn(); // Turn on vibrator when obstacle is detected
            }
        } else {
            buzzerOff(); // Turn off buzzer when no rain or obstacle
            vibratorOff(); // Turn off vibrator when no rain or obstacle
        }

        _delay_ms(10); // Wait for 10 milliseconds before the next reading
    }

    return 0;
}

```